

Math Bootcamp

Day 3 — Remaining real analysis and linear algebra

Math Camp 2026

How to use these slides

Main deck: definitions, recipes, examples, and practice.

Appendix and proofs (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

Not required for Math Camp. Focus on the main slides and exercises unless you want the extra detail.

3.1 Fixed points — equilibrium as a solution

Fixed point. A point x^* is a **fixed point** of a map T if $T(x^*) = x^*$.

Economics reading. Many equilibria are fixed-point problems:

- **Market clearing:** a price p^* with excess demand $E(p^*) = 0$.
- **Game theory:** a strategy profile that is a best response to itself (Nash).

Two useful existence tools (today).

- **Contraction Mapping Theorem:** existence and uniqueness of a fixed point; iteration converges.
- **Brouwer:** continuous map on a closed interval / convex set.

3.1 Contraction Mapping Theorem

Contraction. A map T **shrinks distances** if there is $k \in (0, 1)$ such that $|T(x) - T(y)| \leq k|x - y|$ for all x, y in the domain.

Contraction Mapping Theorem. If $T: [a, b] \rightarrow [a, b]$ is a contraction, then

- **Existence:** there is a fixed point x^* with $T(x^*) = x^*$.
- **Uniqueness:** that fixed point is **unique**.
- **Iteration:** for any starting value $x_0 \in [a, b]$, the sequence $x_n = T(x_{n-1})$ converges to x^* .

3.1 Why iteration converges

Step 1 — Start anywhere. Pick any x_0 . Define $x_n = T(x_{n-1})$ for $n = 1, 2, \dots$

Step 2 — Jumps shrink exponentially. Each new step is at most k times the previous jump:

$$|x_{n+1} - x_n| = |T(x_n) - T(x_{n-1})| \leq k|x_n - x_{n-1}| \leq k^n|x_1 - x_0|.$$

Since $k < 1$, the moves get **exponentially smaller** — the path barely changes after a while.

Step 3 — The sequence converges. Step 2 implies $x_n \rightarrow x^*$ for some limit x^* (check it is a **Cauchy sequence** for a rigorous proof). So iteration from **any** starting point drives x_n toward the same x^* .

Step 4 — The limit is a fixed point. T is continuous, so taking limits in $x_n = T(x_{n-1})$ gives $x^* = \lim x_n = \lim T(x_{n-1}) = T(\lim x_{n-1}) = T(x^*)$.

3.1 Walrasian price adjustment

Market clearing. A price p^* **clears the market** when excess demand is zero: $E(p^*) = 0$. Here $E(p)$ can take **any form** (from supply and demand behind the scenes).

Price adjustment (nonlinear in E). Update the price each period:

$$p_{t+1} = T(p_t) = p_t + \lambda(1 - e^{-\alpha E(p_t)}), \quad \lambda = \frac{1}{5}, \alpha = 0.1.$$

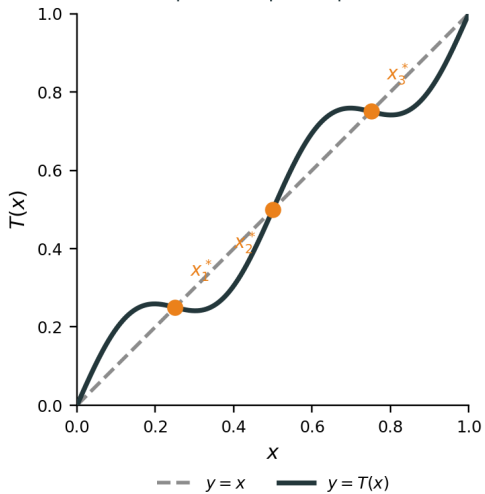
When $|E|$ is small, $1 - e^{-\alpha E} \approx \alpha E$ — move a fraction of excess demand toward balance. Clearing satisfies $E(p^*) = 0 \Leftrightarrow T(p^*) = p^*$. If T is a **contraction** on a price range, then $p_t \rightarrow p^*$ from any starting p_0 .

3.1 Brouwer — one dimension

Brouwer (one dimension). If $T: [a, b] \rightarrow [a, b]$ is **continuous**, then T has **at least one** fixed point x^* with $T(x^*) = x^*$.

Picture. Plot $y = T(x)$ against $y = x$. Any crossing is a fixed point. Continuity forces the graph to meet the diagonal at least once on $[a, b]$.

Multiple fixed points possible



3.1 Brouwer — general statement

Brouwer (several variables). Let $K \subset \mathbb{R}^n$ be **nonempty**, **compact**, and **convex**. If $T: K \rightarrow K$ is **continuous**, then T has at least one fixed point $x^* \in K$ with $T(x^*) = x^*$.

Camp vocabulary.

- **Compact** in $\mathbb{R}^n$: closed and bounded (Heine–Borel).
- **Convex**: the segment between any two points in K stays in K .

One dimension. $K = [a, b]$ is compact and convex, so the 1D case is a special case.

3.1 Brouwer — Cournot duopoly

Two firms. Firm i chooses $q_i \in [0, \bar{q}]$. **Nonlinear** inverse demand $p = a - (q_1 + q_2)^2$ (e.g. $a = 8$, $\bar{q} = 2$).

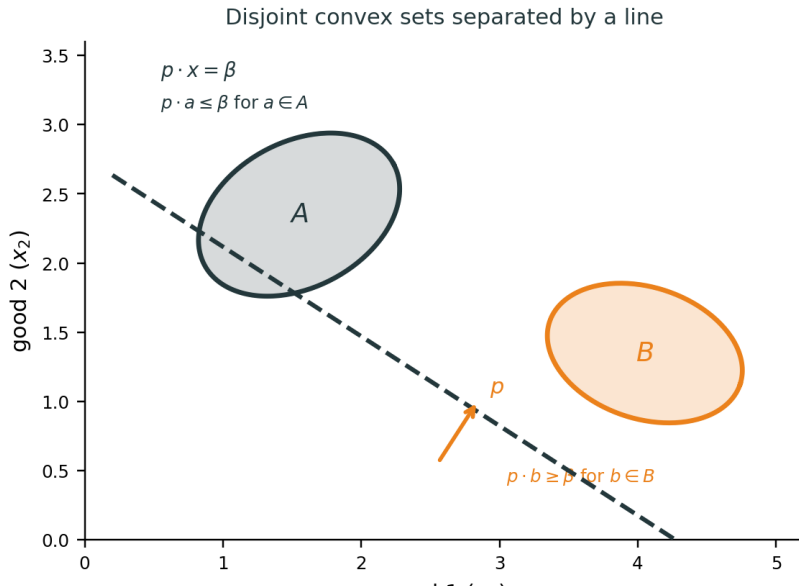
Profit and FOC. $\pi_i = (a - (q_1 + q_2)^2)q_i$. Firm 1's best response solves

$$a - 3q_1^2 - 4q_1q_2 - q_2^2 = 0 \quad (\text{similarly for firm 2 with } q_1, q_2 \text{ swapped}).$$

The map $q_j \mapsto BR_i(q_j)$ is **nonlinear** — no formula like “ $(a - c - q_j)/2$ ”.

Fixed point = Nash. $T(q_1, q_2) = (BR_1(q_2), BR_2(q_1))$ is continuous on $K = [0, \bar{q}]^2$. **Brouwer** guarantees an equilibrium, but you generally solve $T(q_1^*, q_2^*) = (q_1^*, q_2^*)$ by **algebra or numerics** (here one symmetric solution is $q_1^* = q_2^* = \sqrt{a/8}$).

3.2 Separation — picture



3.2 What $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ means

Two dimensions, two goods. Write a bundle as $a = (a_1, a_2)$. A **price line** is

$$p_1x_1 + p_2x_2 = \beta \quad (\text{same as } \mathbf{p} \cdot \mathbf{x} = \beta).$$

Reading the inequalities.

- $\mathbf{p} \cdot a \leq \beta$ for all $a \in A$: every point of A lies on the **same side** of the line (here: on or below it).
- $\mathbf{p} \cdot b \geq \beta$ for all $b \in B$: every point of B lies on the **opposite side** (here: on or above it).

So A and B are **separated** by one straight line; $\mathbf{p}$ is the slope/normal of that line, and β fixes its position. No matrices needed — just prices and a budget line.

3.2 Example — feasible vs. preferred bundles

Setup. One consumer, two goods. Let

- A = **feasible** bundles (can be produced / afforded at given resources);
- B = bundles **strictly preferred** to the status quo (better indifference curves).

If preferences are convex, B is convex. If $A \cap B = \emptyset$, a separating line exists.

Numerical sketch. Suppose the line is $p_1x_1 + p_2x_2 = 100$ with $\mathbf{p} = (2, 5)$.

- Feasible bundle $a = (10, 16)$: $2 \cdot 10 + 5 \cdot 16 = 100$ — on the line (cheapest feasible choice).
- Preferred bundle $b = (14, 18)$: $2 \cdot 14 + 5 \cdot 18 = 118 > 100$ — above the line, not affordable at these prices.

At a competitive allocation, a price line can sit between “what is possible” and “what everyone wants more of.”

3.2 First Welfare Theorem — simple argument

Statement. Under standard convexity, a competitive equilibrium allocation is **Pareto efficient**.

Proof idea (4 steps).

1. In equilibrium, each consumer chooses a best affordable bundle; each firm maximizes profit at prices $\mathbf{p}^*$.
2. Suppose another allocation $\tilde{x}$ is Pareto better: everyone weakly better off, someone strictly.
3. With convex preferences, if $\tilde{x}$ were affordable at $\mathbf{p}^*$, some agent would have chosen it — contradiction.
4. So any Pareto improvement must use bundles that are **not affordable** at $\mathbf{p}^*$: the aggregate “better than” set lies on the expensive side of the equilibrium price line, while feasible aggregates stay on the cheap side. **Separation** is exactly the geometry behind step 4.

3.2 Duality and separation

Primal problem (example). “Choose x to maximize utility subject to a budget / resource constraint.”

$$\max u(x) \quad \text{s.t.} \quad \mathbf{p} \cdot x \leq w.$$

Dual side. The prices $\mathbf{p}$ and cutoff β in $\mathbf{p} \cdot x \leq \beta$ are **shadow values**:

- they tell how much one unit of each good “costs” at the margin;
- in constrained optimization, Lagrange multipliers play the role of $\mathbf{p}$;
- a **separating line** is the same object: coefficients $\mathbf{p}$ and level β that separate feasible from preferred points.

Link. Separation $\Leftrightarrow$ existence of prices supporting an efficient allocation $\Leftrightarrow$ dual variables in the primal problem. Day 4–5 make this algorithmic (LP, KKT).